📘 AI/ML Internship – Day 81 Notes & Tasks

🟢 Module 10: Capstone Project Planning & System Design

📌 Part 1: Recap of Module 9

In the previous module, we learned:

- ✅ Introduction to RAG
- ✅ Document Loaders
- ✅ Text Chunking
- ✅ Embeddings
- ✅ Vector Databases (ChromaDB)
- ✅ Retrievers
- ✅ LangChain
- ✅ Prompt Engineering
- ✅ Multi-PDF RAG
- ✅ Metadata Filtering
- ✅ Conversation Memory
- ✅ RAG Evaluation
- ✅ Deployment

Now it's time to build a complete AI application.

📌 Part 2: What is a Capstone Project?

Definition

A Capstone Project is a final project that combines all the knowledge and skills learned during the internship into one practical application.

Simple Definition

Instead of learning concepts separately, you now build a real AI product that solves a real-world problem.

Example Projects

- AI Resume Screening System
- AI Medical Assistant
- AI Legal Document Assistant
- AI Chatbot using RAG
- AI Student Attendance System
- AI Library Assistant
- AI Invoice Analyzer
- AI Business Forecasting System

📌 Part 3: Why Do We Build a Final Project?

The final project helps interns:

- Apply theoretical knowledge.
- Gain practical development experience.
- Build a portfolio.
- Improve teamwork.
- Prepare for interviews.
- Demonstrate problem-solving skills.

📌 Part 4: AI Project Development Lifecycle

Problem Identification



Requirement Analysis

|

▼

System Design

|

▼

Dataset Collection

|

▼

Model Development

|

▼

Backend Development

|

▼

Frontend Development

|

▼

Testing

|

▼

Deployment

|

▼

Project Presentation

📌 Part 5: Choosing a Project

Before writing code, ask these questions:

- What problem are we solving?
- Who will use the application?
- What data is required?
- Will we use AI, ML, or RAG?
- What are the expected outputs?

Example

Project:

AI Resume Screening System

Problem:

Recruiters spend too much time manually reviewing resumes.

Solution:

AI automatically analyzes resumes and ranks candidates.

📌 Part 6: Requirement Analysis

Requirement analysis defines what the application should do.

Functional Requirements

- User Login
- Upload Files
- AI Prediction
- View Results
- Download Reports

Non-Functional Requirements

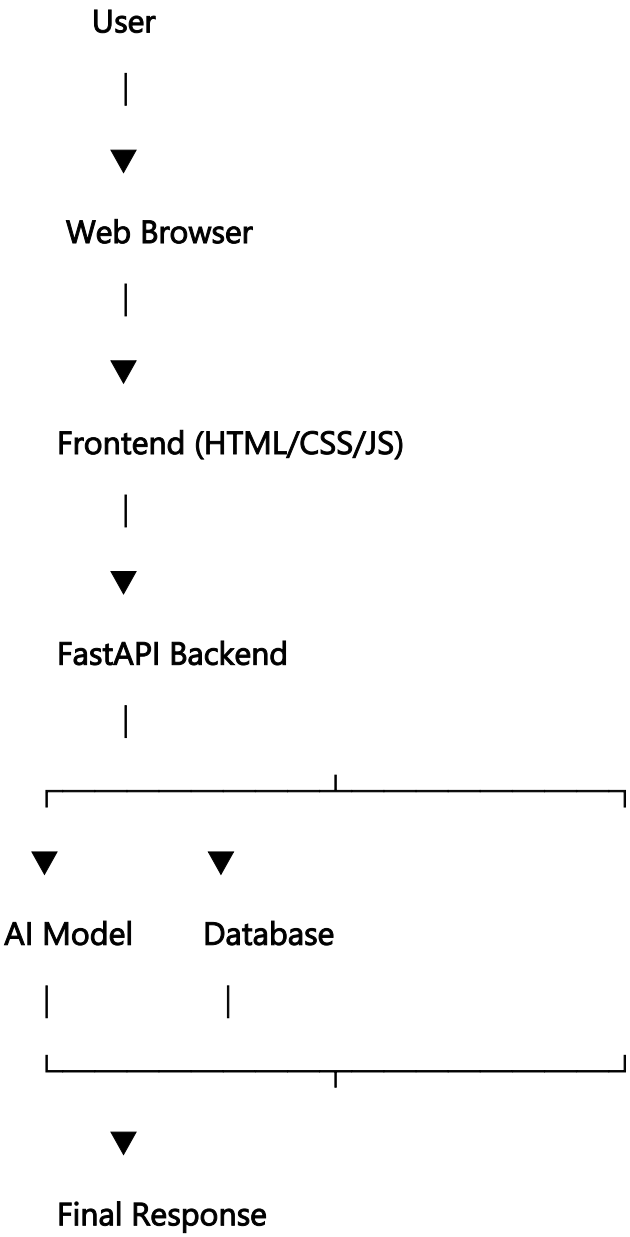
- Fast response
- Secure login

- Easy to use
- Reliable
- Scalable

📌 Part 7: System Design

Before development, create a system design.

Example Architecture:



📌 Part 8: Selecting Technologies

Frontend

- HTML
- CSS
- Bootstrap
- JavaScript

Backend

- FastAPI

Database

- SQLite (Development)
- PostgreSQL / MySQL (Production)

AI Libraries

- LangChain
- Transformers
- Sentence Transformers
- OpenCV (if required)
- Pandas
- NumPy

📌 Part 9: Project Folder Structure

Example:

AI_Project/

```
| —— app/
| —— static/
| —— templates/
```

```
├── models/
├── uploads/
├── chroma_db/
├── database/
├── utils/
├── main.py
├── requirements.txt
└── README.md
```

Keeping the project organized makes it easier to maintain and scale.

📌 Part 10: Team Roles

Role	Responsibility
Project Manager	Plans and monitors the project
AI/ML Engineer	Develops AI models
Backend Developer	Builds APIs and business logic
Frontend Developer	Designs the user interface
Database Engineer	Manages data storage
QA Tester	Tests the application
DevOps Engineer	Deployment and monitoring

📌 Part 11: Best Practices

- ✅ Understand the problem before coding.
- ✅ Divide the project into smaller modules.
- ✅ Use Git for version control.
- ✅ Write clean and readable code.

✅ Test each module separately.

✅ Document the project.

📌 Part 12: Common Mistakes

❌ Starting coding without planning.

❌ Poor folder organization.

❌ Ignoring error handling.

❌ Not testing regularly.

❌ Hardcoding sensitive information like API keys.

📌 Part 13: Real-World Example

Suppose you build an AI Resume Screening System.

Workflow:

User Uploads Resume

|

▼

Resume Parser

|

▼

AI Skill Extraction

|

▼

Candidate Scoring

|

▼

Database



Recruiter Dashboard

This project combines NLP, backend development, databases, and AI.

📌 Part 14: Theory Questions

1. What is a Capstone Project?

A Capstone Project is a final practical project that demonstrates the knowledge and skills learned during a course or internship.

---

2. Why is requirement analysis important?

Requirement analysis helps identify what the application should do before development begins.

---

3. What is system design?

System design defines the architecture, components, and interactions of an application before implementation.

---

4. Name four technologies commonly used in AI projects.

- FastAPI
- LangChain
- ChromaDB
- HTML/CSS/JavaScript

---

5. Why is project planning important?

Project planning reduces errors, improves teamwork, and ensures smooth project execution.

📌 Part 15: Interview Questions

Explain the AI project development lifecycle.

The AI project lifecycle includes problem identification, requirement analysis, system design, data collection, model development, backend and frontend development, testing, deployment, and maintenance.

Why should system design be completed before coding?

It provides a clear roadmap, reduces development errors, and helps all team members understand the architecture.

What are functional and non-functional requirements?

Functional requirements define what the system should do, while non-functional requirements define qualities such as performance, security, and usability.

Why is Git important in software development?

Git helps track code changes, enables collaboration, and allows developers to revert to previous versions if needed.

What makes a good AI project?

A good AI project solves a real-world problem, uses quality data, has a scalable architecture, provides accurate results, and offers a good user experience.

🎯 Tasks for Day 81

🧠 Theory

1. Define a Capstone Project.
2. Explain the AI project development lifecycle.
3. Differentiate functional and non-functional requirements.

4. Draw the architecture of your selected AI project.
5. List the technologies you plan to use.

💻 Practical Task 1

Choose one AI project idea and write:

- Project Title
- Problem Statement
- Objective
- Target Users

💻 Practical Task 2

Design the project architecture using a flowchart or block diagram.

💻 Practical Task 3

Create the project folder structure and initialize a Git repository.

💻 Practical Task 4

Prepare a Software Requirement Specification (SRS) document containing:

- Introduction
- Objectives
- Functional Requirements
- Non-Functional Requirements
- Technologies Used

💻 Practical Task 5

Create a project timeline with estimated milestones for development, testing, deployment, and presentation.